

LESSON 5-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Surface Area of Rectangular Prisms**

Maryland Standard 6.G.A.4 • Learning target – I can find the surface area of a rectangular prism by adding the areas of its six faces.

REMEMBER

- **Surface area** is the total area of all the faces, in **square units**.
- A rectangular prism has 3 pairs of matching faces:
 $SA = 2lw + 2lh + 2wh$.

WORKED EXAMPLE

A box is 8 in $\times$ 3 in $\times$ 2 in. Find its surface area.

Step 1 One face from each pair: $8 \times 3 = 24$, $8 \times 2 = 16$, $3 \times 2 = 6$

Step 2 Add: $24 + 16 + 6 = 46$

Step 3 Each face has a twin, so double: $2 \times 46 = 92$

Answer **92 in²**

WORD BANK

surface area – the total area of all the faces
face – a flat side of a solid

pair – two matching faces
square units – cm², in², ft²

PRACTICE

show your work • label units

1. Find the surface area of the prism.

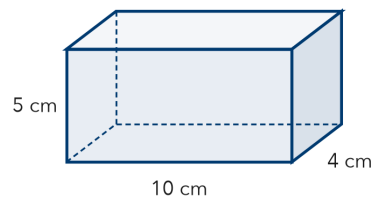
Step 1 Top and bottom: $2 \times (10 \times 4)$.

Step 2 Front and back: $2 \times (10 \times 5)$.

Step 3 Two ends: $2 \times (4 \times 5)$. Then add.

Show your work.

Answer _____ cm²



2. Find the surface area of each solid.

Step 1 Cube: $6 \times (\text{edge} \times \text{edge})$.

Step 2 Prism: $2lw + 2lh + 2wh$.

a. cube, edges 3 cm _____ cm²

b. cube, edges 10 ft _____ ft²

c. prism 4 m $\times$ 2 m $\times$ 1 m _____ m²

d. prism 5 in $\times$ 5 in $\times$ 2 in _____ in²

e. cube, edges 5 in _____ in²

f. prism 6 cm $\times$ 3 cm $\times$ 2 cm _____ cm²

3. **Gift Box.** Nadia wraps a box. It is 12 in long, 8 in wide, and 4 in tall. Her paper covers 400 in². Is that **enough** paper? Explain.

Step 1 Find the surface area of the box.

Step 2 Compare it to 400 in².

Show your work.

Answer _____

4. **Toy Chest.** A toy chest is 4 ft long, 2 ft wide, and 3 ft tall. Leo paints all faces but the bottom. How many ft² does he paint?

Step 1 Find the top only (no bottom).

Step 2 Find the front and back, then the two sides.

Step 3 Add the 5 faces.

Answer _____ ft²

5. **Find the Error.** Diego says the surface area of a cube with 2 cm edges is 8 cm² because $2 \times 2 \times 2 = 8$. Explain his mistake and find the surface area.

Frame Diego found the ___, not the surface area. A cube has ___ faces, so ___.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Pick a box at home. Point to each pair of matching faces and tell someone how you would find its surface area.

LESSON 5-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Surface Area of Triangular Prisms**

Maryland Standard 6.G.A.4 • Learning target – I can find the surface area of a triangular prism using its net.

REMEMBER

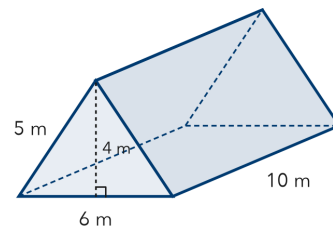
- A triangular prism has **2 triangle bases** and **3 rectangles**.
- Each rectangle is one side of the triangle $\times$ the prism's length. Add all **5 faces**.

WORKED EXAMPLE

Prism 10 cm long. Ends are right triangles with sides 3, 4, 5 cm.

Step 1 Triangles: $2 \times (\frac{1}{2} \times 3 \times 4) = 12$ **Step 2** Rectangles: $30 + 40 + 50 = 120$ **Step 3** Add: $12 + 120 = 132$ **Answer** **132 cm²****WORD BANK****triangular prism** – 2 triangle ends + 3 rectangles**rectangle face** – triangle side $\times$ prism length**height** – straight up from the base, at a square corner**surface area** – add all 5 faces**PRACTICE***show your work • label units*

1. Find the surface area. The triangle ends have sides 5 m, 5 m, and 6 m.

Step 1 Two triangles: $2 \times (\frac{1}{2} \times 6 \times 4)$.**Step 2** Rectangles: each side $\times$ 10.**Step 3** Add all 5 faces.*Show your work.***Answer** _____ m²

2. Find each area.

a. 2 triangles, base 8 ft, height 3 ft: _____ ft²

b. 1 rectangle, 5 ft by 9 ft: _____ ft²

3. **Tent.** A tent is a **triangular prism**, 9 ft long. Each end is a triangle: base 8 ft, height 3 ft, and two 5 ft sides. Find the fabric for **all 5 faces**, with the floor.

Step 1 Find the two ends.**Step 2** Find the floor and the two sides.**Step 3** Add.**Answer** _____ ft²

- 4. Doorstop.** A wooden doorstop is a triangular prism 4 in long. Its ends are right triangles with sides 5 in, 12 in, and 13 in. Find its surface area.

Step 1 Ends: the legs 5 in and 12 in are the base and height.

Step 2 Rectangles: 5×4 , 12×4 , 13×4 .

Step 3 Add all 5 faces.

Answer _____ in²

- 5. Find the Error.** For the prism in Exercise 1, Lucas found the rectangles as 6×10 , 4×10 , and 4×10 . Explain his mistake.

Frame Lucas used the triangle's ____ instead of its ____ for two rectangles.

TEST PRACTICE

choose the best answer

- 6.** Triangle ends: 9 cm² each. Rectangles: 20, 20, and 24 cm². What is the surface area?
A 64 cm² **B** 73 cm² **C** 82 cm² **D** 146 cm²
- 7.** A prism is 8 cm long. Its triangle ends have sides 6, 8, and 10 cm. What is the total area of the 3 rectangles?
A 24 cm² **B** 192 cm² **C** 240 cm² **D** 480 cm²

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____